World Action Model

从视频生成先验到机器人动作

Purshow's Notes

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World Action Model：从视频生成先验到机器人动作

UniPi

Learning Universal Policies via Text-Guided Video Generation

将控制严格分成视觉规划与动作解码两步。

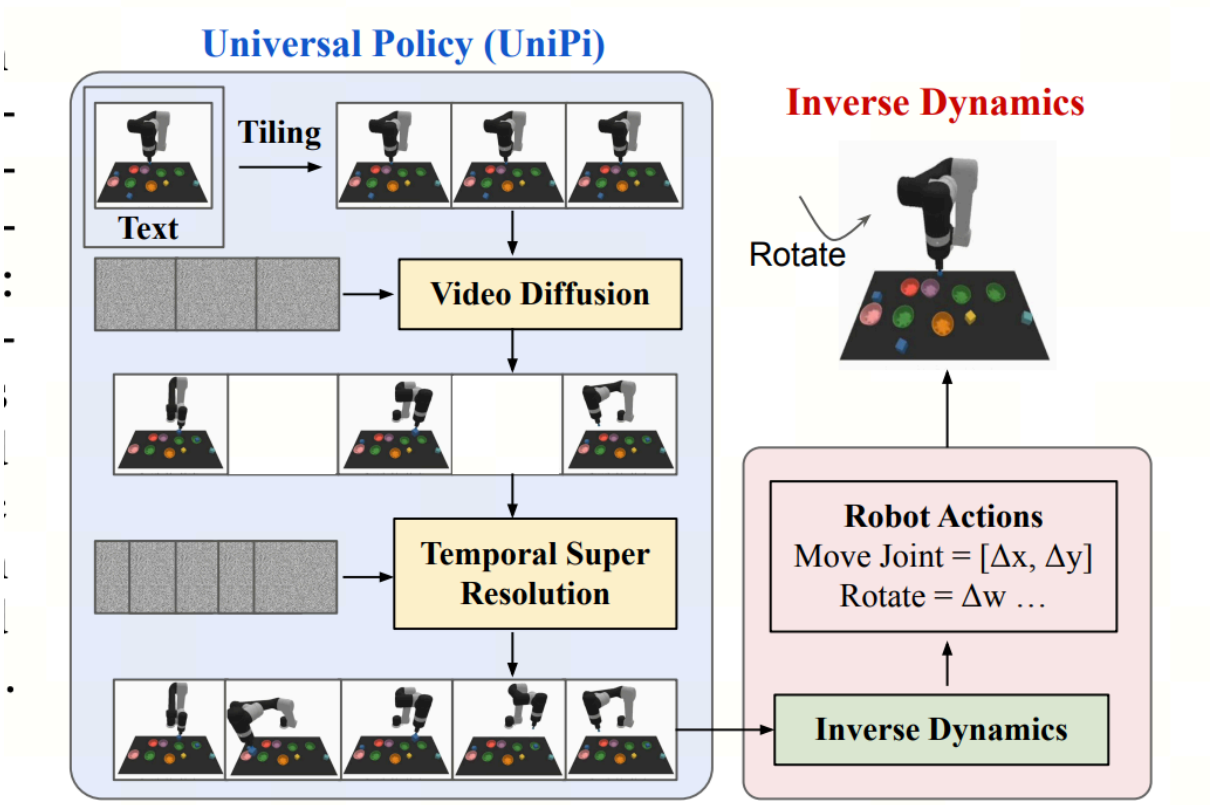


Figure 2: Given an input observation and text instruction, we plan a set of images representing agent behavior. Images are converted to actions using an inverse dynamics model.

$(o_t, l) \rightarrow$ text-guided video diffusion $\rightarrow \hat{o}_{t+1:t+K} \rightarrow$ inverse dynamics $\rightarrow a_{t:t+H}$.

训练视频规划器时可以使用模拟轨迹、真实机器人视频和 YouTube 等无动作视频，因为视频损失只要求观测序列；真正的机器人动作 grounding 则由带动作数据训练的 IDM 完成。推理时必须先生成未来 RGB 计划，视频不是辅助损失，而是动作产生所需的中间变量。优点是计划可视化且规划器不绑定单一本体；限制是视频生成会有错误、时间对齐误差和 IDM 误差会串联传播，完整视频去噪也带来显著开销。

PAD

Prediction with Action: Visual Policy Learning via Joint Denoising Process

不再采用“先视频、后 IDM”的串联方式，而是在同一个 DiT 中同时去噪未来图像和动作：

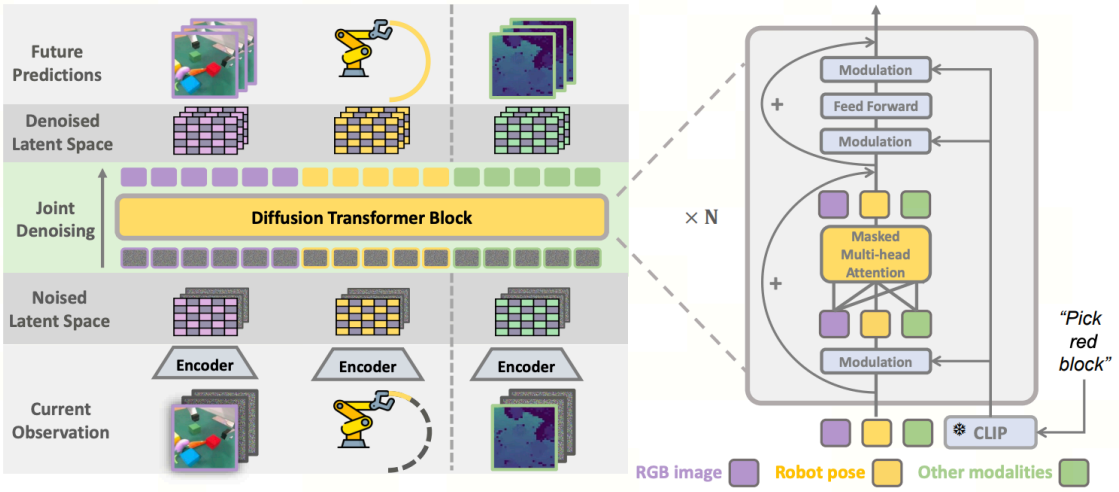


Figure 3: Visualization of the PAD framework. Current observations in different modalities are first encoded into latent and concatenated with white noise channel-wise. These noised latent are then tokenized into tokens and perform a joint denoising process to predict the images and robot actions simultaneously. PAD can flexibly accommodate extra or missing modal inputs through a masked-attention mechanism

$(o_t, s_t, l, \epsilon_{\text{video}}, \epsilon_a) \rightarrow$ joint DiT denoising $\rightarrow (\hat{o}_{t+1:t+K}, \hat{a}_{t:t+H})$.

带动作机器人轨迹同时监督两种输出；无动作视频可以把动作 token 替换为 mask，只更新视觉预测部分，并通过共享 transformer 间接影响动作表示。推理时模型仍执行联合未来预测和动作生成，动作和视频共享 backbone。

UWM

Unified World Models

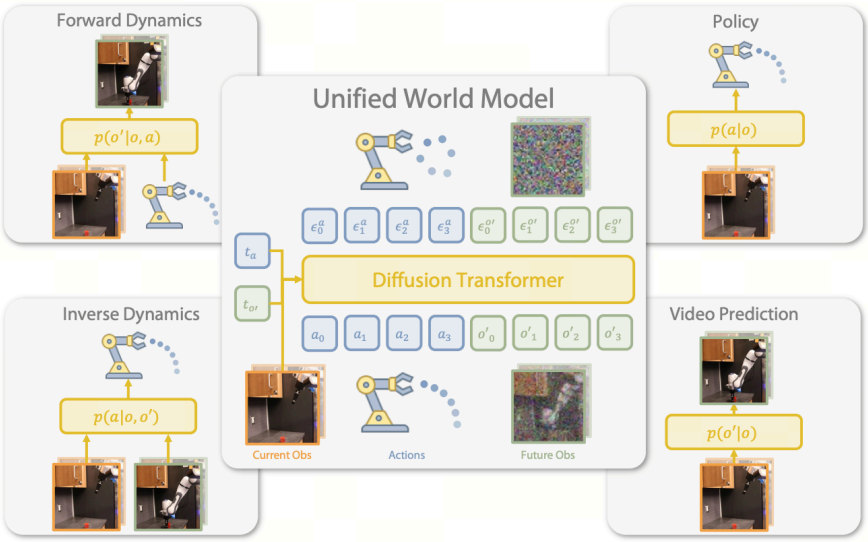


Fig. 1. Unified World Models integrates action and video diffusion in a unified transformer architecture controlled by modality-specific diffusion timesteps. The model can be trained on large robotics datasets and then flexibly perform a variety of different inferences at test time. This enables improved robustness and generalization for imitation learning.

把视频和动作放进同一扩散过程，并且为二者使用独立 diffusion timestep。统一模型学习联合分布 po, a, o'，通过调节两个模态的噪声级别，分别查询：

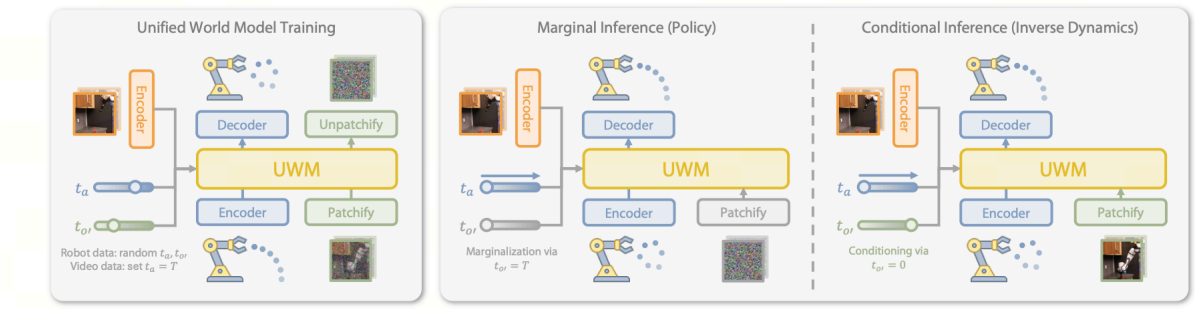


Fig. 2. Unified World Model training and inference pipeline. The left panel shows UWM pretraining on robot trajectories with actions and co-training on action-free videos by masking out actions using diffusion timesteps. The right panel illustrates marginal and conditional inference modes, corresponding to the policy and the inverse dynamics.

$$p(a \mid o), \quad p(o' \mid o, a), \quad p(a \mid o, o'), \quad p(o' \mid o).$$

策略推理时，未来图像的扩散时间固定在最大噪声 T，相当于将其边缘化，只对动作去噪。无动作视频则反过来把动作保持在最大噪声，只训练 action，推理时候也不需要推理 video。论文实验中的 action-free 数据主要来自忽略动作标签的机器人视频，而不是大规模普通互联网视频。

UVA

Unified Video Action Model

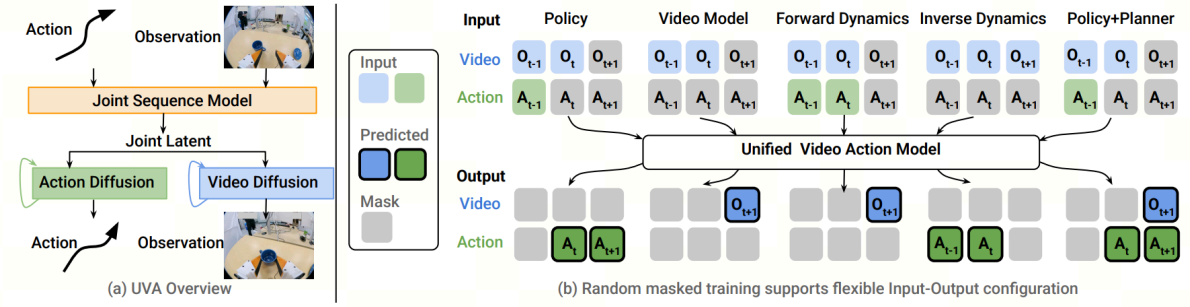


Fig. 1: **Unified Video Action Model.** (a) UVA features a *joint* video-action latent representation and *decoupled* video-action decoding. The joint latent representation effectively models the underlying relationships between video and action sequences, while the decoupled diffusion enables high-speed action inference by bypassing video generation. (b) By leveraging masked training, UVA supports flexible input-output combinations for actions and videos. This versatility allows the model to function as a robot policy, video model, forward or inverse dynamics model, or even a combined policy and video planner—all within a unified framework.

mask 魅力时刻 again，masked training 让模型可充当 policy、video generator、forward dynamics 或 IDM。标准 policy 模式直接从共享 latent 调用 action head，不需要借助 video gen。与 UWM 不同的是 UWM 是在 diffusion 过程和主干参数上联合；UVA 在 shared latent representation 上联合，但在输出解码时分开使用两个 mar head，这样 action diffusion 的时候会更快。

VPP

Video Prediction Policy

采用两阶段训练。第一阶段用互联网人类操作、互联网机器人操作和下游机器人视频，将 Stable Video Diffusion 微调为文本条件 Manipulation TVP；第二阶段冻结 TVP：

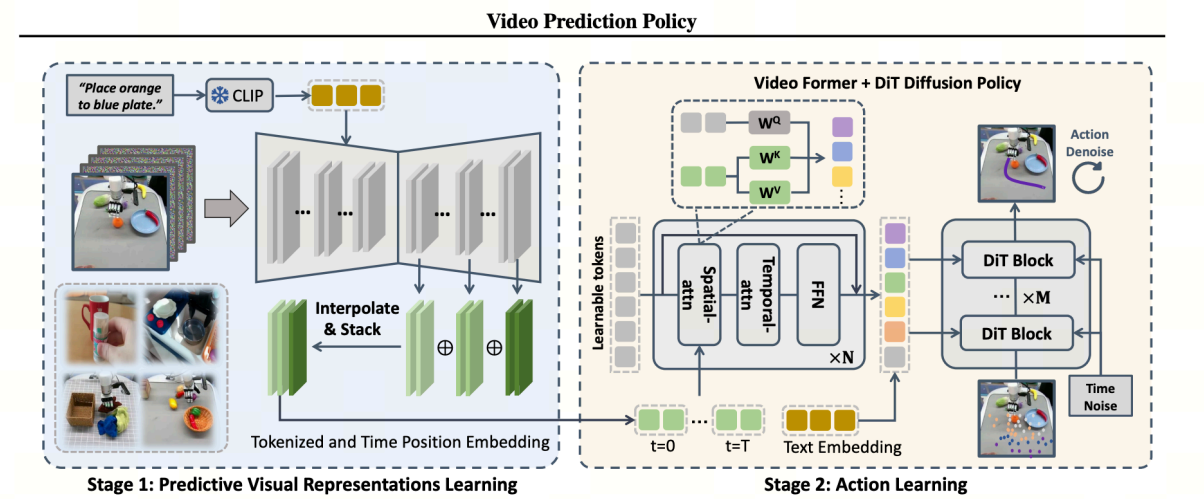


Figure 2. In the first stage, VPP fine-tunes a general-purpose video foundation model into a manipulation-focused Text-guided Video Prediction (TVP) model using robot and internet manipulation datasets. In the second stage, we use video-former to aggregate the representations from the TVP model during the first forward pass, followed by the diffusion policy head. This approach enables VPP to learn an implicit inverse dynamics model from the predicted future while maintaining a high control frequency.

$(o_t, l, \epsilon_{\text{future}}) \rightarrow$ TVP 单次前向 $\rightarrow h_{\text{predictive}} \rightarrow$ Video-Former $\rightarrow$ diffusion action head.

直接从第一次前向过程的中间层提取 predictive visual representation。动作 head 相当于一个隐式 IDM。

从这里开始进入现代 ***WAM***

mimic-video

*mimic-video: Video-Action Models for Generalizable Robot Control Beyond VLAs *

这篇文章我很喜欢

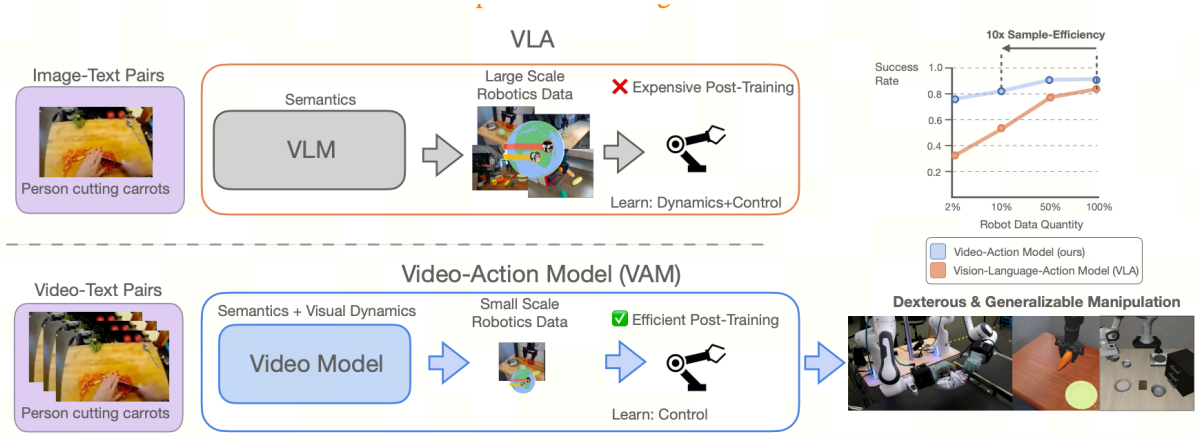


Fig. 1: We introduce mimic-video, a new class of Video-Action Model (VAM) that grounds robotic policies in pretrained video models. Unlike standard VLAs that must learn physical dynamics from scratch (top), mimic-video leverages the inherent visual dynamics of video backbones to isolate the control problem (bottom). This enables state-of-the-art performance on dexterous manipulation tasks, while achieving 10x greater sample efficiency compared to VLAs (right).

经典起手式之：VLA 不行！

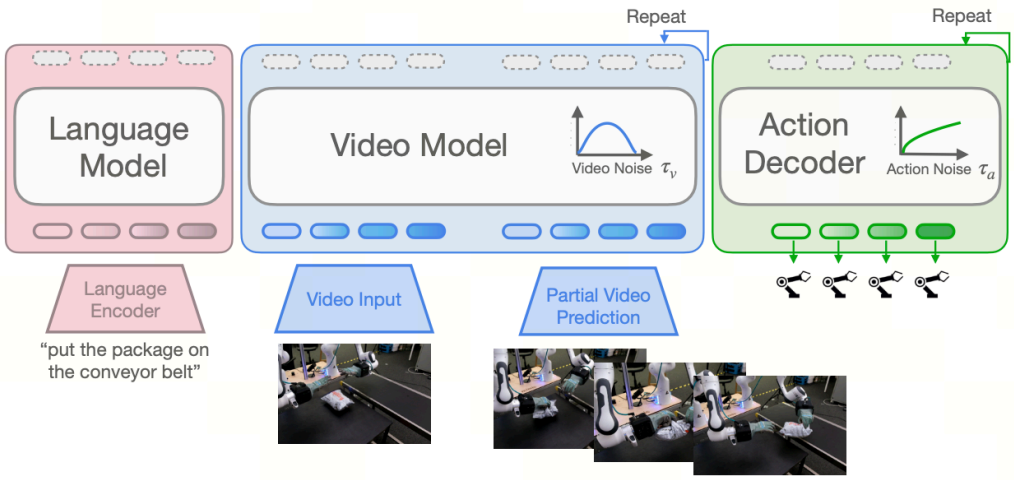


Fig. 3: mimic-video architecture: we instantiate our framework with a pretrained video generation backbone (Cosmos-Predict2 [38, 37]), which provides rich physical dynamics priors learned from large-scale video data. We adapt this model for control via a *partial denoising* strategy, where the video backbone follows the flow to an intermediate flow time τ_v to extract latent visual plans. These representations condition a smaller action decoder, which processes proprioceptive states and predicts action trajectories. The video and action components operate on independent flow schedules (τ_v and τ_a), allowing us to design the learning problem separately for each modality.

$(o_{\text{past}}, l, \epsilon_{\text{future}}) \rightarrow$ 部分视频 flow $\rightarrow h^{\tau_v}_{\text{video}} \rightarrow$ action flow/IDM $\rightarrow a_{t:t+H}$.

- 视频骨干: 预训练的视频扩散模型 Cosmos-Predict2, 在机器人视频上 LoRA 微调;
- 动作解码器: 轻量 flow-matching 解码器, 当 IDM, 条件于视频骨干的中间 latent 表征;
- 训练两阶段:① LoRA 微调视频骨干;② 冻结骨干、从零训动作解码器。
- 部分去噪：选择高噪声 step 去噪时候的中间层表征，类似 VPP，原因可以看 appendix D，很值得品下，简单记下我的理解：

    1. 缓解训推误差（有没有想到 rae 和 diffusion forcing hhh）
    2. 高噪是正则
    3. 高噪时候的模型是表征好的时候，低噪的时候只是 refine
    4. Action policy 需要的不是像素质量最高的视频，而是对动作最有用的视频生成内部表征。

DreamZero

World Action Models are Zero-shot Policies

dreamzero 似乎是 WAM 的命名之作，现在 WAM 有这样的热度，有一部分原因也归功于 Jim Fan 的力推

我觉得 dreamzero 开场提到的 wam 优势是值得被深入探讨的：

1. 从多样化、非重复数据中高效学习，
2. 开放世界泛化，
3. 仅从视频数据中进行跨具身学习，
4. 对新机器人进行 few-shot 适应

做法是使用自回归视频 DiT，在单一模型中联合流匹配视频 latent 与动作。

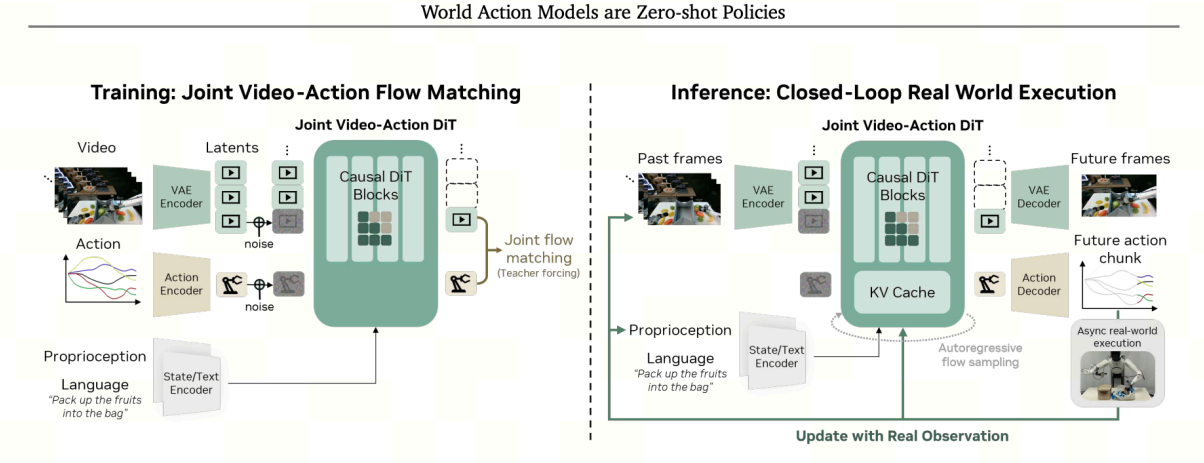


Figure 4: **Model Architecture of DreamZero.** The model takes three inputs: visual context (encoded via a VAE), language instructions (via a text encoder), and proprioceptive state (via a state encoder). These are processed by an autoregressive DiT backbone using flow matching, which jointly predicts future video frames and actions through separate decoders. During training (left), for each chunk, the model denoises noisy video and action latents conditioned on clean video context. During inference (right), predictions are executed asynchronously in the real world, and ground-truth observations are fed back into the KV cache to prevent error accumulation.

$$\underbrace{\pi_0(\mathbf{o}_{l:l+H}, \mathbf{a}_{l:l+H} \mid \mathbf{o}_{0:l}, \mathbf{c}, \mathbf{q}_l)}_{\text{DreamZero}} = \underbrace{\pi_0(\mathbf{o}_{l:l+H} \mid \mathbf{o}_{0:l}, \mathbf{c}, \mathbf{q}_l)}_{\text{video prediction}} \underbrace{\pi_0(\mathbf{a}_{l:l+H} \mid \mathbf{o}_{0:l+H}, \mathbf{q}_l)}_{\text{IDM}}$$

即视频预测与 IDM 的端到端组合，共享时间步，联合去噪。推理时，每个 chunk 生成未来视觉表示和对应动作；动作执行后，预测帧在 KV cache 中被真实观测替换，以限制长期误差积累（这个操作很典，后面 lingbot-va 也会用）。DreamZero-Flash 解耦了噪声调度，让动作完成去噪时，当前 chunk 的视频表示仍可处于较高噪声状态，从而加速。

此外就是一些结论：WAM 体现了一定的 OOD 泛化；多样性比重复性更重要；视频 backbone 存在明显规模效应；AR 与双向架构平均分相同但 AR 动作更平滑、支持 KV cache、推理快 3–4 倍，并更容易保持语言—视频—动作时间对齐；视频-only 跨 embodiment 数据有效；机器人通常忠实执行视频分支的计划，主要失败来自错误的视频规划，而不是动作提取失败；人类视频数据能提高未见任务表现。

当然，dreamzero 由于是发版模型 tech report，一些情况下消融可能很难以做的非常非常完美无缺。

Fast-WAM

Fast-WAM Do World Action Models Need Test-time Future Imagination?

Fast-WAM 它专门检验“训练时视频预测”与“推理时未来想象”是否必须绑定，有用的是训练表征，还是将推理时候的视频。训练时，未来视频 latent 和动作都接受 flow-matching 损失，但注意力掩码禁止 action token 看到 future video token，并且在推理时直接出 action。我觉得这是一篇很棒的学术论文，尽管他研究的思想可能在于之前的论文里面就有提及，但我觉得具身智能缺少很多篇这种类型的论文，去具体探讨一个现象究竟是否存在和到底是怎么样的机理。

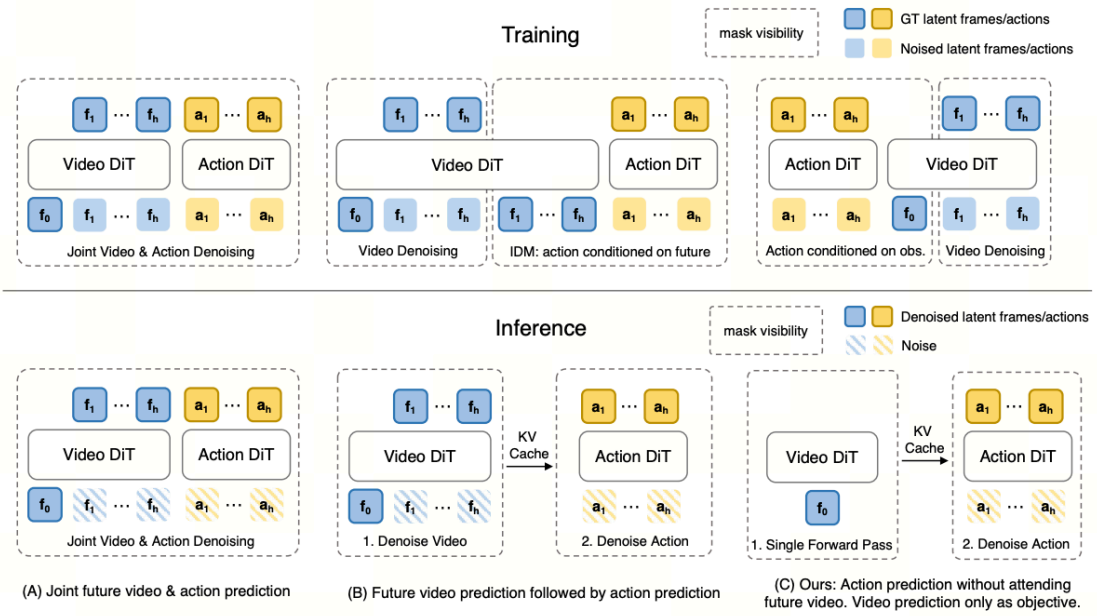
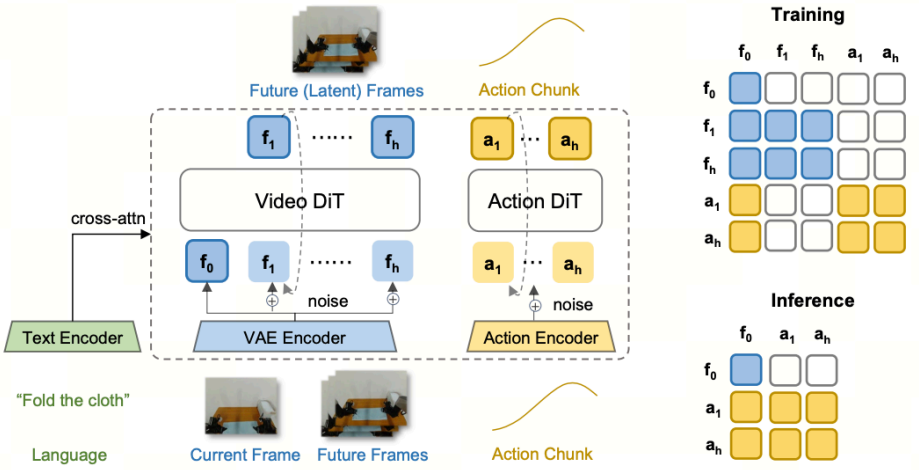


Figure 1: Three representative WAM paradigms. (A) Joint-modeling WAMs denoise future video and action tokens together. (B) Causal WAMs first generate future observations and then condition action prediction on the generated future representation. (C) Fast-WAM retains video co-training during training but removes explicit future generation at inference time, directly predicting actions from latent world representations in a single forward pass.



(a) Fast-WAM model architecture.

(b) Training and inference masks.

Figure 2: Fast-WAM architecture and the structured attention mask used to disentangle video co-training from action generation.

论文的对比显示，在其评测设置中，删除训练期视频目标的损害大于删除测试期想象，当然，可能是受限于资源，这只能证明显式未来并非文章中的仿真任务上的必要条件，还不能证明所有任务都不需要 video rollout。

Motus

Motus: A Unified Latent Action World Model

用的是现在全世界无论是 unify 界还是具身界都最在用的 Mixture-of-Transformers（MoT）连

接理解、视频生成和动作三个 expert，并通过视频、动作各自独立的时间步实现五种推理模式：

VLA， WM， IDM， VGM， Video-Action Joint.

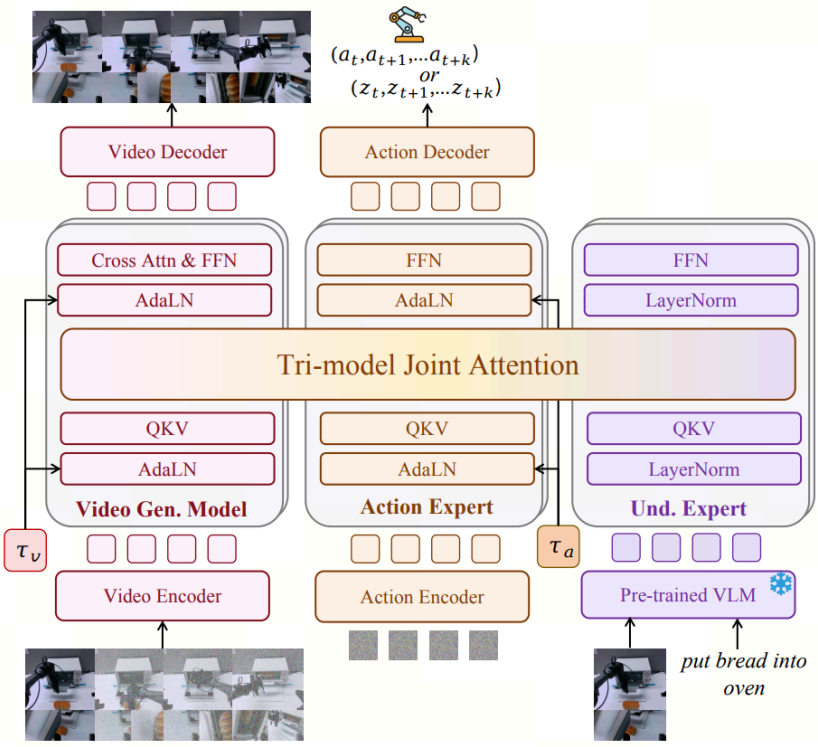


Figure 1. Motus Architecture. Here, $a_t \ldots a_{t+k}$ are actions, $z_t \ldots z_{t+k}$ are latent actions, and τ_v and τ_a are the rectified flow timesteps for the video generation model and the action expert, respectively.

Motus 还通过 optical flow 训练了 latent action vae 压缩“delta action”，以跨数据源学习运动先验。这个操作其实会跟之后我们讲的 lapa adaworld 以及 repwam 有些相似，实际上都是想给无 action 标签的视频通过重建打一个伪标签方便模型学习.

Cosmos Policy

Cosmos Policy: Fine-Tuning Video Models for Visuomotor Control and Planning

还记得最早大家做 video gen，其实最喜欢的就是画 world model 的大饼：如果做了什么，世界模型会怎样。

Cosmos Policy 就是还记得最初的梦想，在 predict 未来的同时还是 predict value，评估这个动作的价值

具体而言，它不增加独立 action transformer，将机器人 action、proprioception 和 value 归一化后复制、reshape 成与视频 VAE latent 相同形状的“伪 latent frame”，再与当前图像、未来图像 latent 拼成统一序列，通过 latent frame injection 微调 Cosmos-Predict2：

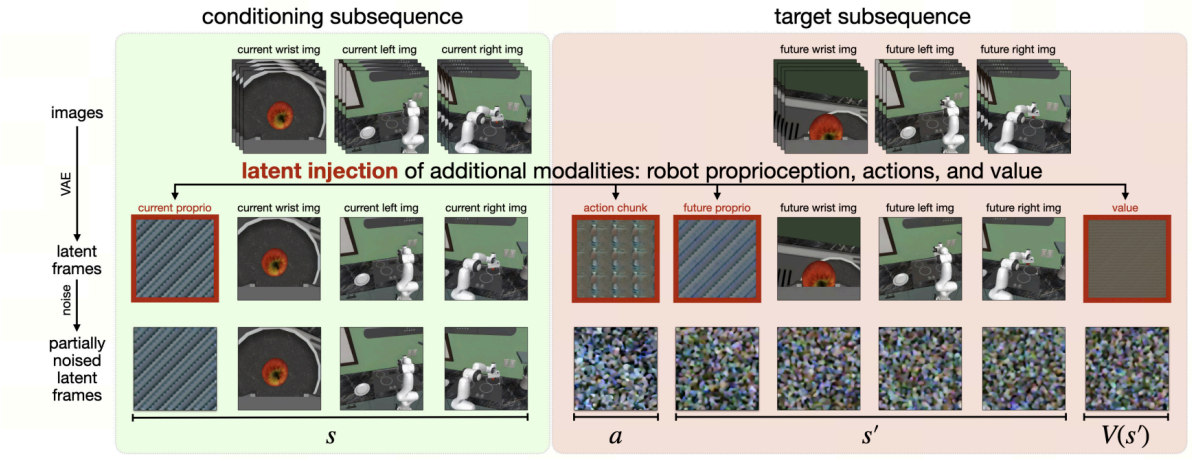


Figure 2: The latent diffusion sequence of Cosmos Policy. We illustrate *latent frame injection*—the primary mechanism for adapting the pretrained Cosmos-Predict2 into a policy that can predict robot actions, future states, and values without architectural changes. First, raw images are tokenized into latent frames (first row). Then, additional modalities are inserted directly into the latent frame sequence of the video diffusion model (second row). The model is then tasked to denoise the noised latent frames conditioned on the clean frames (third row). See Section 4.1 for more details. (Note: For simplicity, this figure does not depict certain implementation details; see Figure 8 for a more detailed visualization.)

训练时仅通过不同的 conditioning mask，让同一个 diffusion backbone 分别学习 $p(a, s', V \mid s)$（policy）、$p(s', V \mid s, a)$（world model）和 $p(V \mid s, a, s')$（value）。推理时既可直接生成 action，也可用 **Best-of-N**：采样多个 action → world model 想象对应未来状态 → value model 打分 → 选择价值最高的 action

LingBot-VA

Causal World Modeling for Robot Control

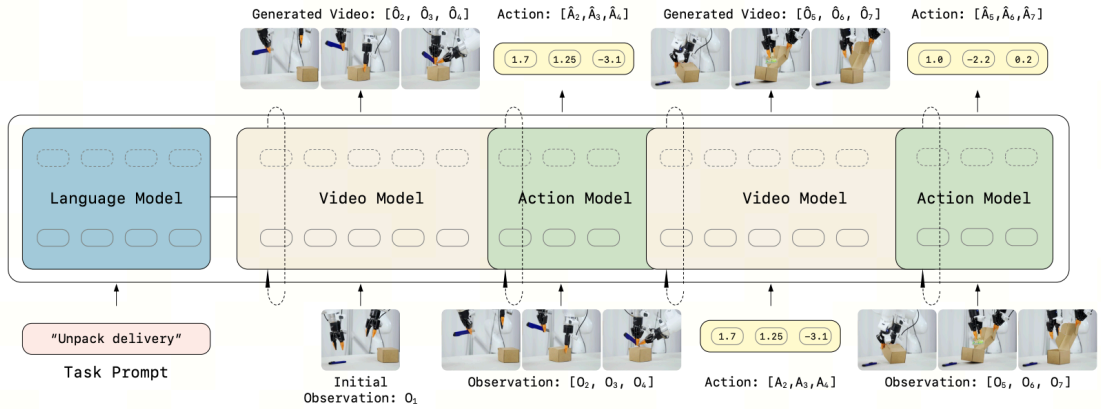


Figure 2. Framework overview: LingBot-VA is conditioned by *autoregressive diffusion* for unified *video-action world modeling*. We leverage a dual-stream *Mixture-of-Transformers (MoT)* architecture that interleaves video and action tokens within a single sequence. At each autoregressive step, the video stream (initialized from Wan2.2-5B) first predicts future latent visual states via *flow matching*. Then the action stream decodes corresponding actions through *inverse dynamics* conditioning on the predicted visual transitions.

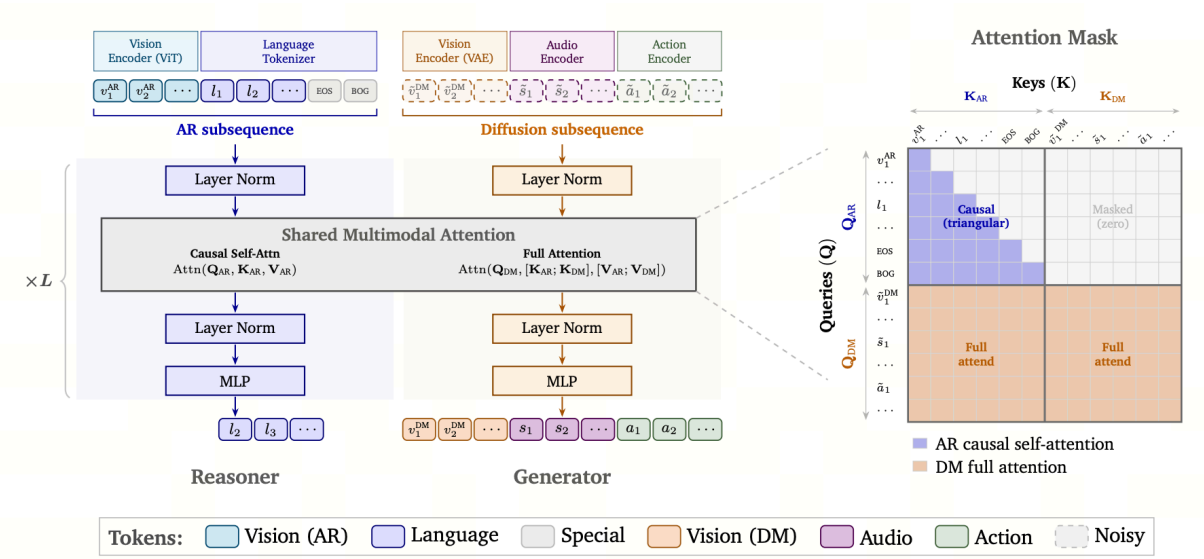
伟大的集大成之作啊：

chunk 自回归，MoT，下采样视觉（每 4 个动作时刻，只保留一个视觉时刻）；teacher forcing 训练推理执行后重新观察真实世界更新，，Noisy History Augmentation 让模型更鲁棒，视觉 latent 可以只去噪一部分（3 步 / 0.6），action 维度低交互 attention 的时候映射到 video ()

1. 相机以较低频率提供视觉
   每个视觉状态对应 4 个高频动作
   ↓
2. 真实历史视频 + 历史动作进入 KV cache
   ↓
3. Video stream 从噪声预测未来视频 latent
   只跑 3 步，到 s=0.6
   ↓
4. 部分去噪的未来视频 token
   与 action token 做联合 attention
   action 768 → 3072 → attention → 768
   ↓
5. Action stream 完整去噪到 s=1
   输出可执行动作
   ↓
6. 机器人执行动作
   ↓
7. 相机重新观察真实世界
   用真实视觉替代长期生成历史
   ↓
8. 开始下一轮

Cosmos3

Cosmos 3: Omnimodal World Models for Physical AI



是的！还是 MoT！还是联合去噪！

值得再聊聊 action encoder，是一堆 domain-specific Linear Projection，而不是平常一个共享的 input MLP。

先将不同 embodiment 的动作统一表示为 ego pose / effector pose / grasp state，再通过各 embodiment 独立的 Linear 映射过去。

Qwen-RobotWorld

Qwen-RobotWorld Technical Report

3.1 Model Architecture

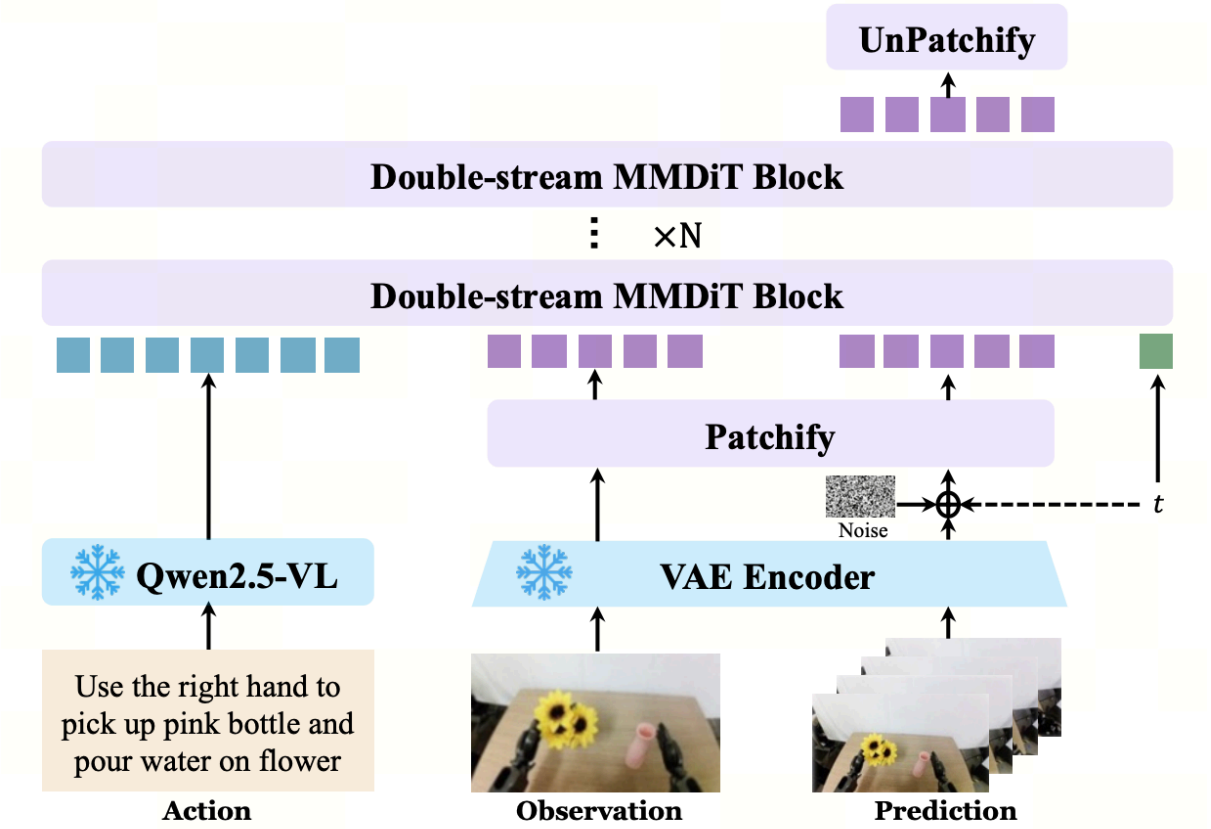


Figure 3: Overview of our video generation architecture with 60-layer double-stream MMDiT backbone.

Qwen-Image 的 Robo 场景的 Video Gen 版

把自然语言当成统一的 Action Space：不管是机械臂关节动作、自动驾驶控制还是导航指令，都转换成语言，然后统一建模为**"当前视觉状态 + 语言动作 → 未来视频"**。模型使用 60 层 Double-Stream MMDiT，一路输入冻结的 Qwen2.5-VL 提取的动作语义，另一路输入 VAE 的视频 latent，并通过逐层 joint attention 融合。数据上构建 EWK，包含约 860 万视频文本对、2 亿 + 帧、20+ 种 embodiment 和 500+ 动作类别；训练则先用 T2I/T2V/TI2V 学 general world prior，再持续混入 general data 的同时逐步增加单视角、多视角、复杂 manipulation、driving/navigation 等 embodied 数据进行 specialization。

Motusbrain

Motubrain: An Advanced World Action Model for Robot Control

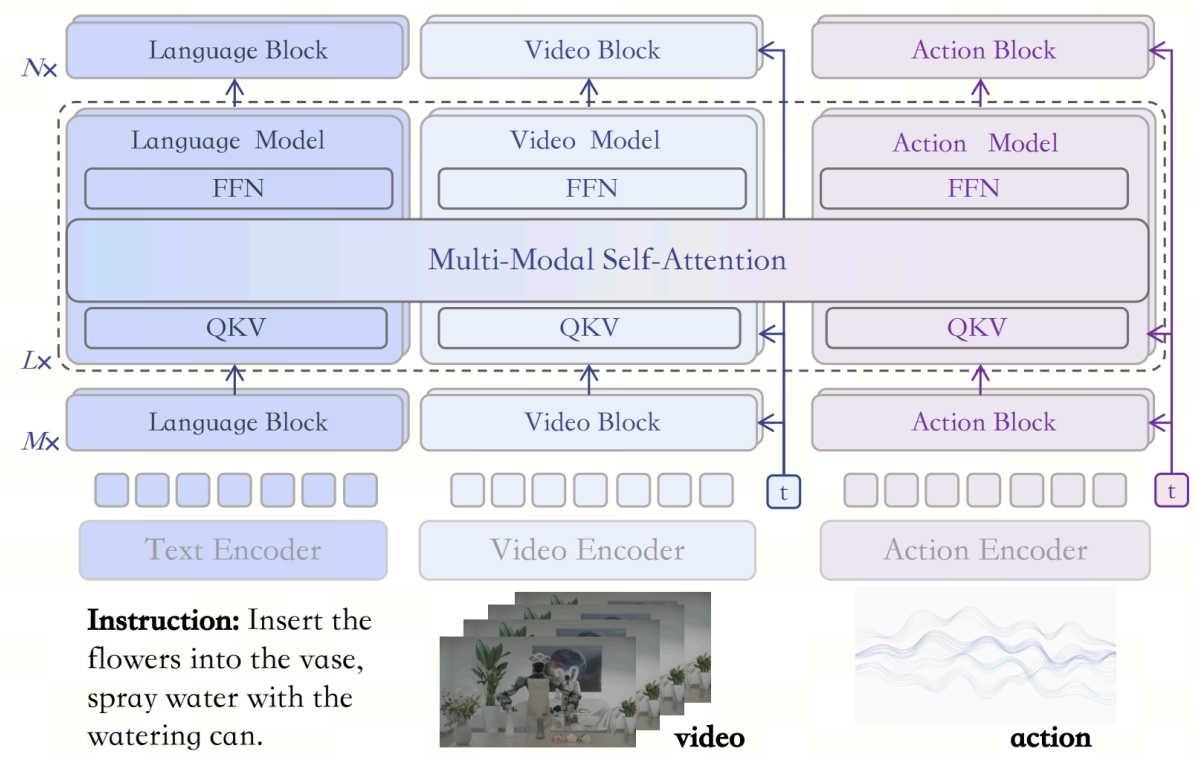


Figure 1 Overview of `Motubrain`'s architecture. `Motubrain` builds on a unified video-action backbone and adopts a three-stream Mixture-of-Transformers architecture with text, video, and action streams. It further uses an H-bridge attention design to balance cross-modal interaction and efficiency, while supporting flexible multiview inputs through view-dependent 3D RoPE offsets.

MoT again again. 不同的是为了降低跨模态计算量，只在中间 50% Transformer 层进行 Video–Action joint attention（H-Bridge），其余层保持 modality-specific。同时支持 VLA policy、world model、video generation、IDM 和 video-action joint prediction。模型从 Vidu 视频生成模型初始化，先用 egocentric + heterogeneous robot data 训练 video/world branch，再冻结 video branch、训练 action branch；不同机器人统一到以当前末端执行器状态为参考的 relative EEF action，每个 EEF 用 position + 6D rotation + gripper 共 10 维表示。随后针对目标机器人做 Non-AR 或 chunk-level AR post-training，并采用 V2A asymmetric attention，使 action 可以看 video/text。

τ0-WM

τ0-WM: A Unified Video-Action World Model for Robotic Manipulation

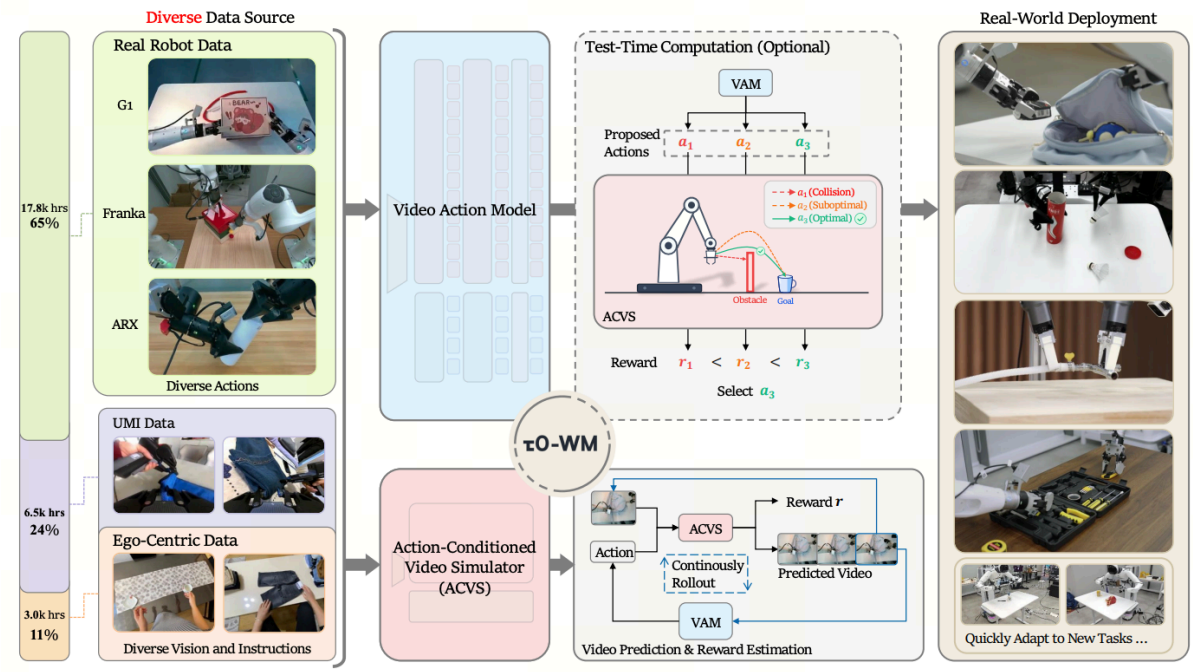


Fig. 1. **Overview of the τ_0-WM framework.** Heterogeneous interaction data from real robots, UMI-style collection, and egocentric human videos are used to train a Video Action Model and an Action-Conditioned Video Simulator. At deployment, the system proposes action candidates, evaluates imagined futures through test-time computation and simulator-based scoring, and selects or rectifies actions for robust manipulation across tasks and embodiments.

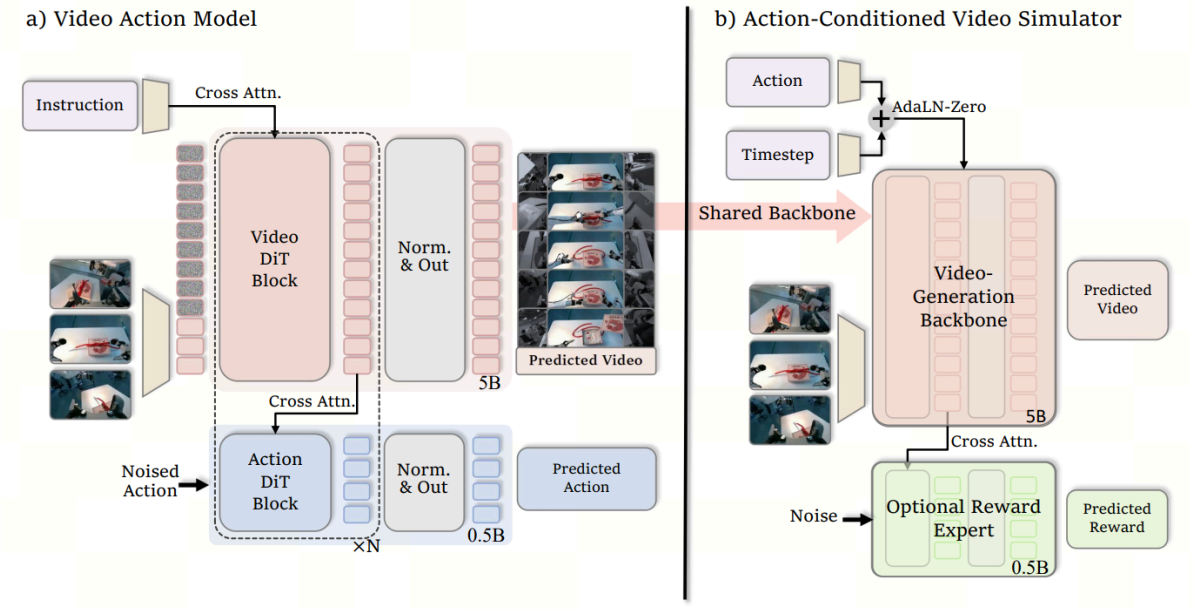


Fig. 2. **Architecture of τ_0-WM.** The Video Action Model (VAM) serves as the policy interface, jointly predicting future visual latents and executable action chunks with a shared video backbone and an Action DiT branch coupled through cross-attention. The Action-Conditioned Video Simulator (ACVS) serves as the evaluation interface, reusing the video-generation backbone to roll out VAM-proposed action chunks and predict dense reward scores for test-time action selection.

推理时，VAM 首先根据当前 observation、instruction 和 robot state，从不同 Gaussian noise 出发采样多个 action candidates。随后使用 **RCS（Re-denoising Consistency Score）**对这些动作做低成本筛选：将每个候选 action 重新加噪到某个中间 flow timestep，再送回同一个 VAM 的 Action DiT，预测对应的 flow velocity，并与理论目标的 v 计算 MSE；RCS 定义为该误差的负值，RCS 越高，说明该 action 位于 VAM 学到的 action manifold 上越稳定、越可信。（这个 claim 很有意思，不过还是需要进一步的验证之类的，尤其是与成本的 trade-off）

此外还有个一个约 0.5B 的 Reward Expert 通过 cross-attention 读取 Video DiT 的中间特征，预测 dense reward trajectory。对多个 action 分别得到对应的 future 和 reward 后，选择 reward 最高

的未来；必要时再将这个高 reward future 作为条件送回 VAM，重新生成更合适的 action 后执行。

想法上非常非常有趣，当然，是否 work 还是需要更完整的消融与更大规模的 scale.

LDA-1B

LDA-1B: Scaling Latent Dynamics Action Model via Universal Embodied Data Ingestion

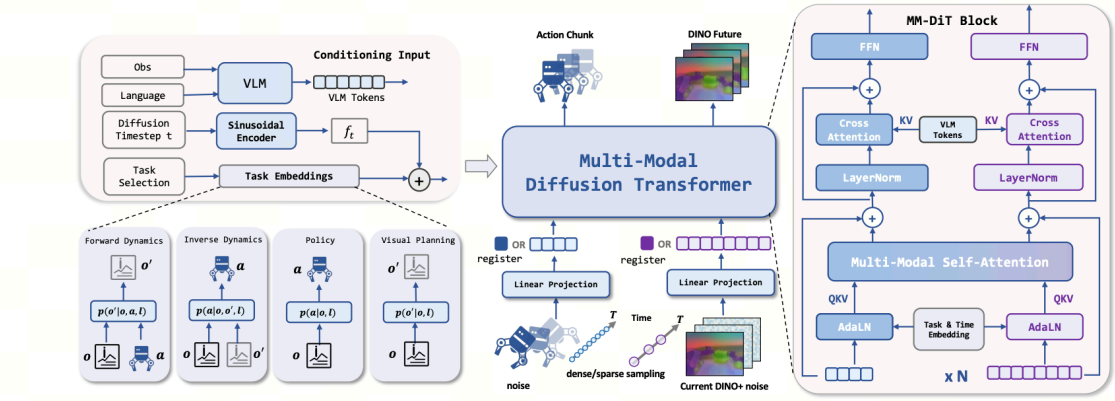


Fig. 2: **Architecture of LDA.** LDA jointly denoises action chunks and future visual latents under multiple co-training objectives, including policy learning, forward dynamics, inverse dynamics, and visual forecasting. Conditioned on VLM tokens, diffusion timesteps, and task embeddings, the model adopts a multimodal diffusion transformer architecture, where action and visual experts are decoupled and interact through a shared self-attention layer.

MM-DiT！

一个 claim 的是各种数据都是有用的，高质量轨迹同时训练 policy 和 dynamics；低质量/次优轨迹不直接教 policy，而主要训练 forward dynamics / visual forecasting；没有 action label 的人类视频只训练 visual forecasting。未来视觉状态也不在 pixel/VAE latent 中预测，而是在冻结的 **DINOv3 feature space**（还是那句话，真希望有一天真能出现一个很棒的 for robo 的 encoder 哇，当然，这个事情目前来看可能会是自上而下，先有一个 general 表征，再垂到 robo teak 上的）中预测。模型主体是约 1B MM-DiT：action token 和 DINO visual token 各自拥有 modality-specific QKV/FFN，但进入共享 attention 相互交互，Qwen3-VL 提供语言/当前视觉语义条件；action 以 10 Hz、visual 以 3 Hz 建模，二者通过 Flow Matching 同时进行去噪预测。

LingBot-VA 2.0

Native Video-Action Pretraining for Generalizable Robot Control

其实我已经反复看几遍了，但是这次写 blog 还是忍不住想真吊啊真吊啊，要么现在只能看一篇 WAM，那绝对是 LingBot-VA2。

1. RepWAM 的 Encoder
2. native 因果生成（我太爱这个了），Next Forcing 用来 MTP
3. action-video from scratch pretrain
4. MoE 基模（基建应该是得力于 LingBot-Video，也是非常棒的作品）
5. 甚至赠送 In-context learning

下面我们来具体聊下 repwam 和 next-forcing

Action Encoder 讲 repwam 前，我们先讲一些前置工作，包包饺子

LAPA *Latent Action Pretraining from Videos*

这篇并不是 wam，但是这篇的思想影响蛮多的。

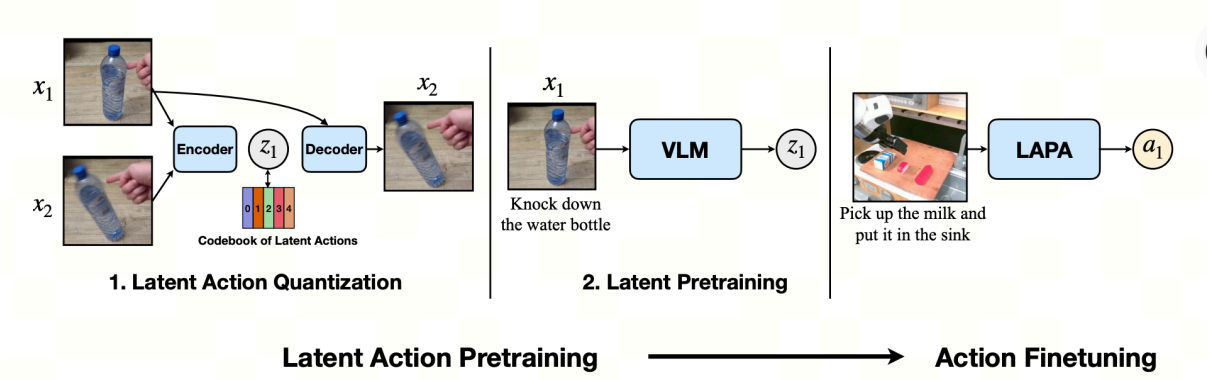


Figure 2: **Overview of Latent Action Pretraining**. (1) Latent Action Quantization: We first learn discrete latent actions in a fully unsupervised manner using the VQ-VAE objective (Detail in Figure 8). (2) Latent Pretraining: The VLM is trained to predict latent actions, essentially performing behavior cloning. After pretraining, we finetune LAPA on a small set of action-labeled trajectories to map the latent space to the end effector delta action space.

简而言之，就是训练了一个离散 vq 码本，目标是解决“互联网视频没有机器人动作标签，无法直接用于 VLA 预训练“的问题。它先用前后视频帧训练一个 latent action tokenizer，把视觉变化压缩成离散的动作 token；再让 VLM 根据当前图像和语言指令预测这些 token，从而学习物体交互和动作先验；最后在少量机器人数据上替换为真实 action head 并整体微调。

AdaWorld *AdaWorld: Learning Adaptable World Models with Latent Actions*

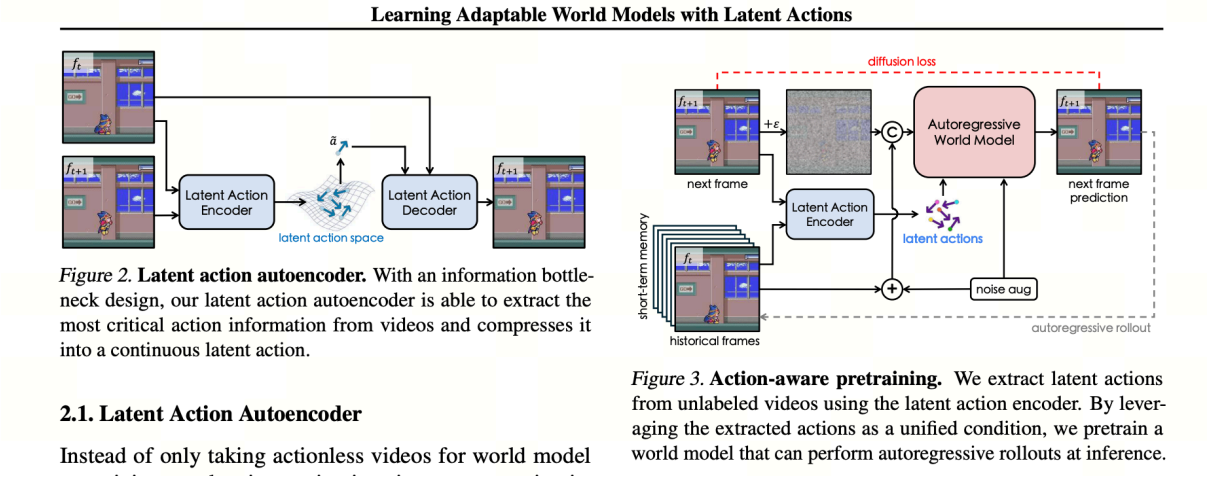


Figure 2. **Latent action autoencoder.** With an information bottleneck design, our latent action autoencoder is able to extract the most critical action information from videos and compresses it into a continuous latent action.

2.1. Latent Action Autoencoder

Instead of only taking actionless videos for world model

Figure 3. **Action-aware pretraining.** We extract latent actions from unlabeled videos using the latent action encoder. By leveraging the extracted actions as a unified condition, we pretrain a world model that can perform autoregressive rollouts at inference.

跟 LAPA 类似，但是从 vq 换成了为世界模型准备的 VAE，从相邻视频帧中提取连续 latent action，使该变量描述“当前状态如何变化到下一状态”；再训练 diffusion world model，根据历史帧和 latent action 生成未来帧，后面作者团队应该是将这套上到了 DreamDojo，非常棒的工作。

RepWAM *RepWAM: World Action Modeling with Representation Visual-Action Tokenizers*

饺子包完了，现在开始讲醋

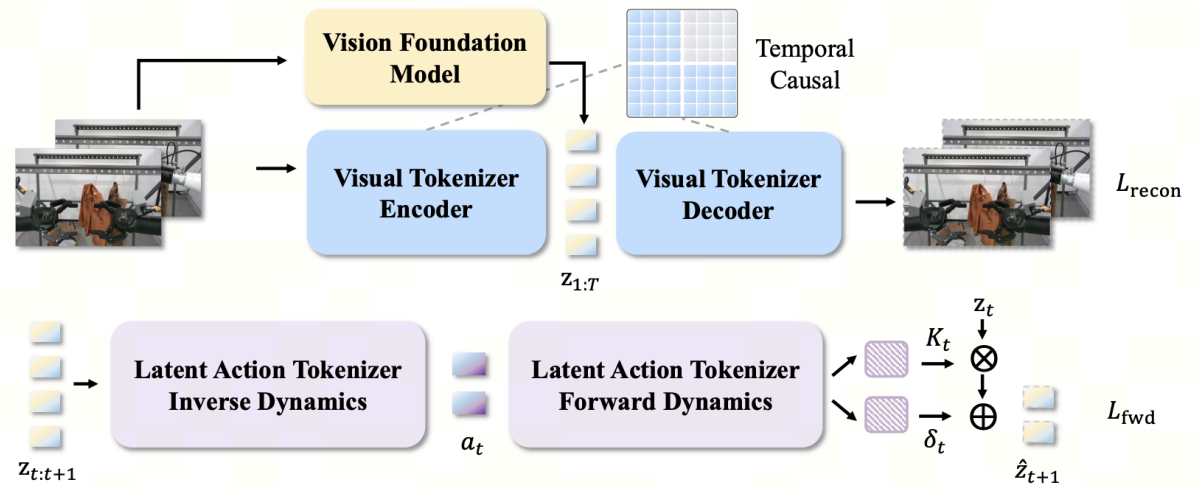


Figure 1 Overview of our representation visual-action tokenizer, which aligns visual latents with a frozen visual foundation model to obtain semantically rich visual tokens, and induces latent actions as transitions within this shared semantic space via coupled inverse and forward dynamics models.

Again, RepWAM 重新设计了 WAM 的 visual/action latent space。首先训练一个 RepViTok semantic visual tokenizer：仍然采用 ViT Autoencoder，并结合像素重建损失、perceptual loss 和 GAN loss 保证视频重建质量，同时额外将 video latent 对齐到冻结的视觉基础模型 Perception Encoder 特征，使 latent 不只保留纹理，还显式包含 object identity、spatial relation 等高层语义。随后在该 semantic visual latent 上训练 Latent Action Tokenizer（LAT）：IDM 根据相邻两帧的 visual latent，将“状态如何发生变化”压缩成一个 4 维 latent action；FDM 再根据当前 visual latent 和 latent action，预测 token 之间的 transport 关系以及 residual，从而重建下一时刻的 latent。

Table 3 Comparison between different VAEs for the 1.3B WAM on RoboTwin 2.0 simulation, Easy and Hard, 50 tasks.

Metric	WAN2.2 VAE [24]		RepViTok (Ours)	
	Easy	Hard	Easy	Hard
$\text{Average}_{\text{hor=1}}$	81.1	78.4	**86.2**	**83.1**
$\text{Average}_{\text{hor=2}}$	75.5	73.9	**85.7**	**84.0**
$\text{Average}_{\text{hor=3}}$	67.2	68.0	**92.0**	**85.4**
$\text{Average}_{\text{50 Tasks}}$	78.0	76.0	**86.6**	**83.1**

Table 4 Ablation study on latent actions for world action modeling. We compare w/o latent actions, joint prediction, and two-stage latent action training.

Model	Eval Seen				Eval Unseen				PickFruit
	gFVD	PSNR	SSIM	OLS	gFVD	PSNR	SSIM	OLS	
w/o	61.01	18.47	0.70	18.82	72.91	17.72	0.67	14.15	30%
Joint Pred	94.25	15.24	0.61	18.52	98.77	15.09	0.55	15.22	20%
Two Stages (Ours)	**48.23**	**22.86**	**0.75**	**19.87**	**58.83**	**19.93**	**0.74**	**16.98**	**50%**

我一向是非常喜欢这类表征 encoder 类的工作的，我觉得是一个非常好的方向，如何得到一个重建/可学性/压缩率 tradeoff 好的 encoder 此事在图片生成早有记载，并且我也之前写过一篇 blog 讲述它的演进，尽管我最后叛逃选择了去除 encoder(。

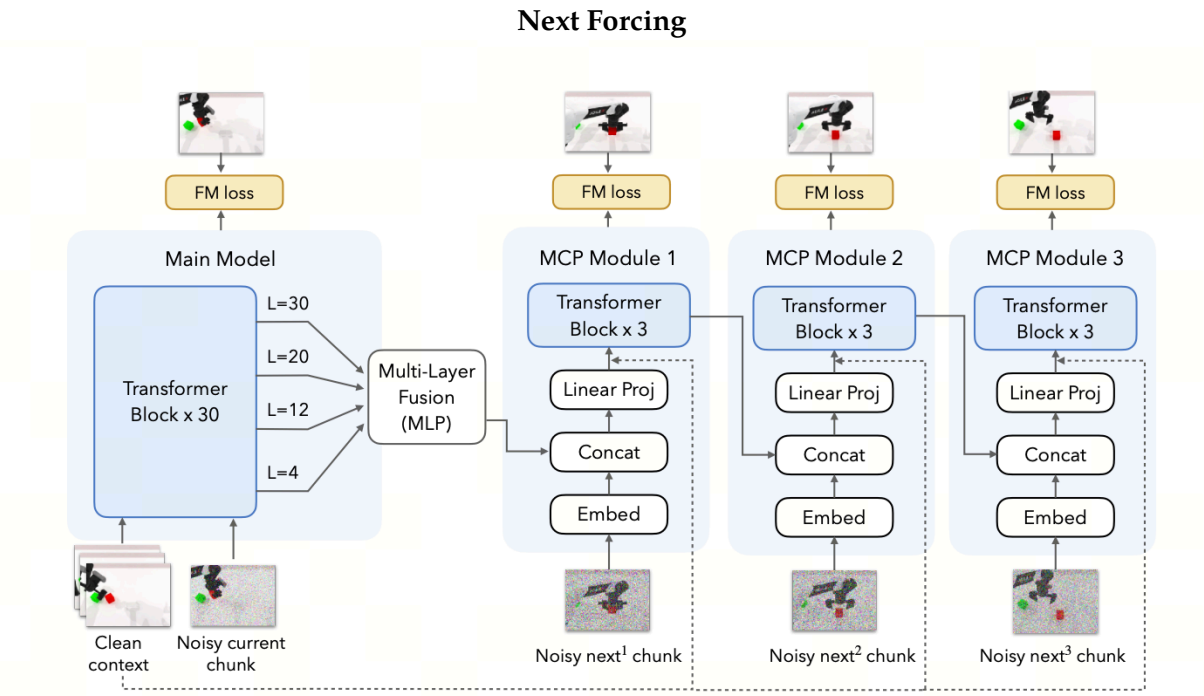


Figure 2: **Overview of Next Forcing.** The main model denoises the current chunk, while chained MCP modules predict future chunks (next^1, next^2, . . .) using features from the main model, providing dense temporal supervision during training and enabling parallel chunk prediction at inference.

Next Forcing 认为传统 WAM 的 teacher forcing 只监督“当前 next chunk”，尤其在高帧率下相邻 chunk 几乎一样，模型很容易通过“复制上一帧外观 + 小修正”完成 denoising，而不必真正学习。(此事 streaming video gen 亦有记载)

为此，它在 LingBot-VA 的 video stream 上增加 3 个轻量 **Multi-Chunk Prediction（MCP）** 模块，让主模型预测当前 chunk 的同时，额外预测 `next`^{1} / `next`^{2} / `next`^{3} 三个未来 chunk。具体来说，未来 chunk 各自独立加噪并做 Flow Matching；MCP 使用比主模型更大的 timestep shift（默认 $s_{\text{main}} = 5$, $s_{\text{mcp}} = 10$），故意把未来 target 加得更 noisy，使 MCP 更依赖主干提供的 temporal representation；同时取主模型第 $\{4, 12, 20, 30\}$ 层 hidden states，经 MLP 融合后送入 MCP，并让三个 MCP 形成 `next1` → `next2` → `next3` 的 causal chain。整体目标为

$$\mathcal{L} = \mathcal{L}_{\text{video}} + \mathcal{L}_{\text{action}} + \sum_{k=1}^{3} w_k \mathcal{L}^k_{\text{MCP}}.$$

默认 $w = (0.5, 0.2, 0.1)$。MCP 本身只作用于 video stream，action stream 仍沿用 LingBot-VA 的 inverse-dynamics Flow Matching，通过 MoT 中的跨模态交互间接受益。推理时 MCP 可以全部扔掉，实现与原模型相同 inference cost；也可以保留第一个 MCP，让主模型生成当前 chunk 的同时 MCP 并行生成下一个 chunk。